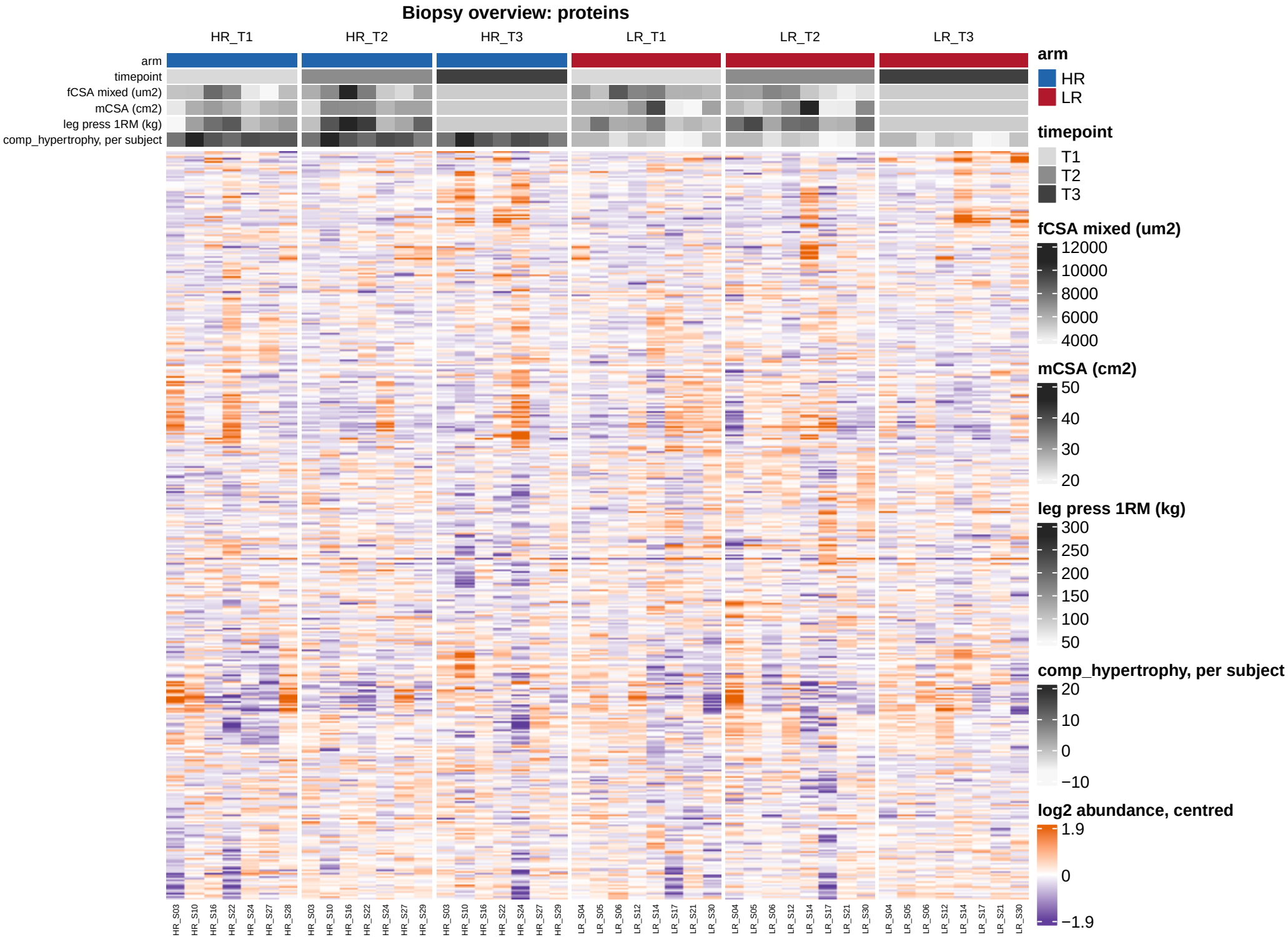
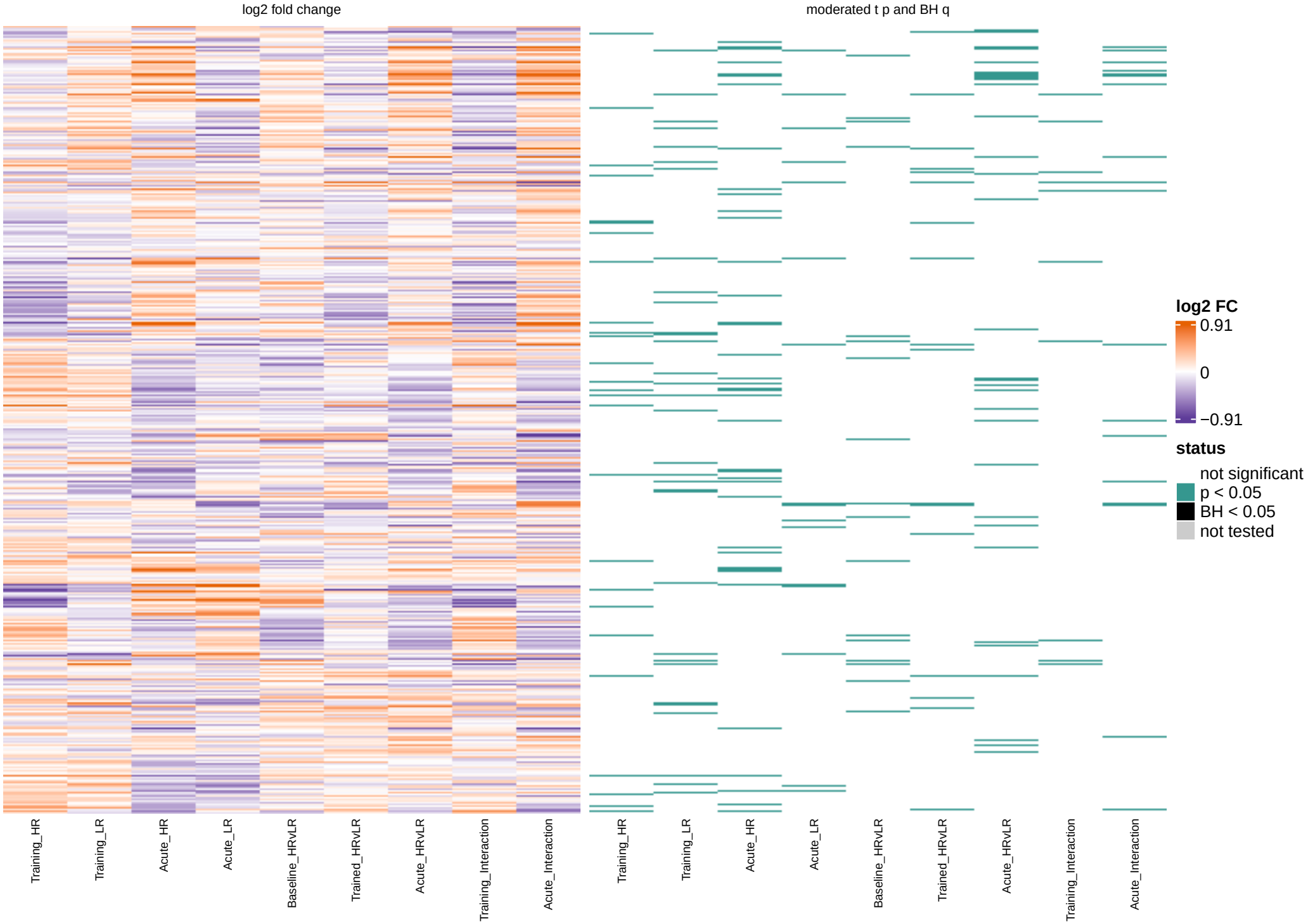
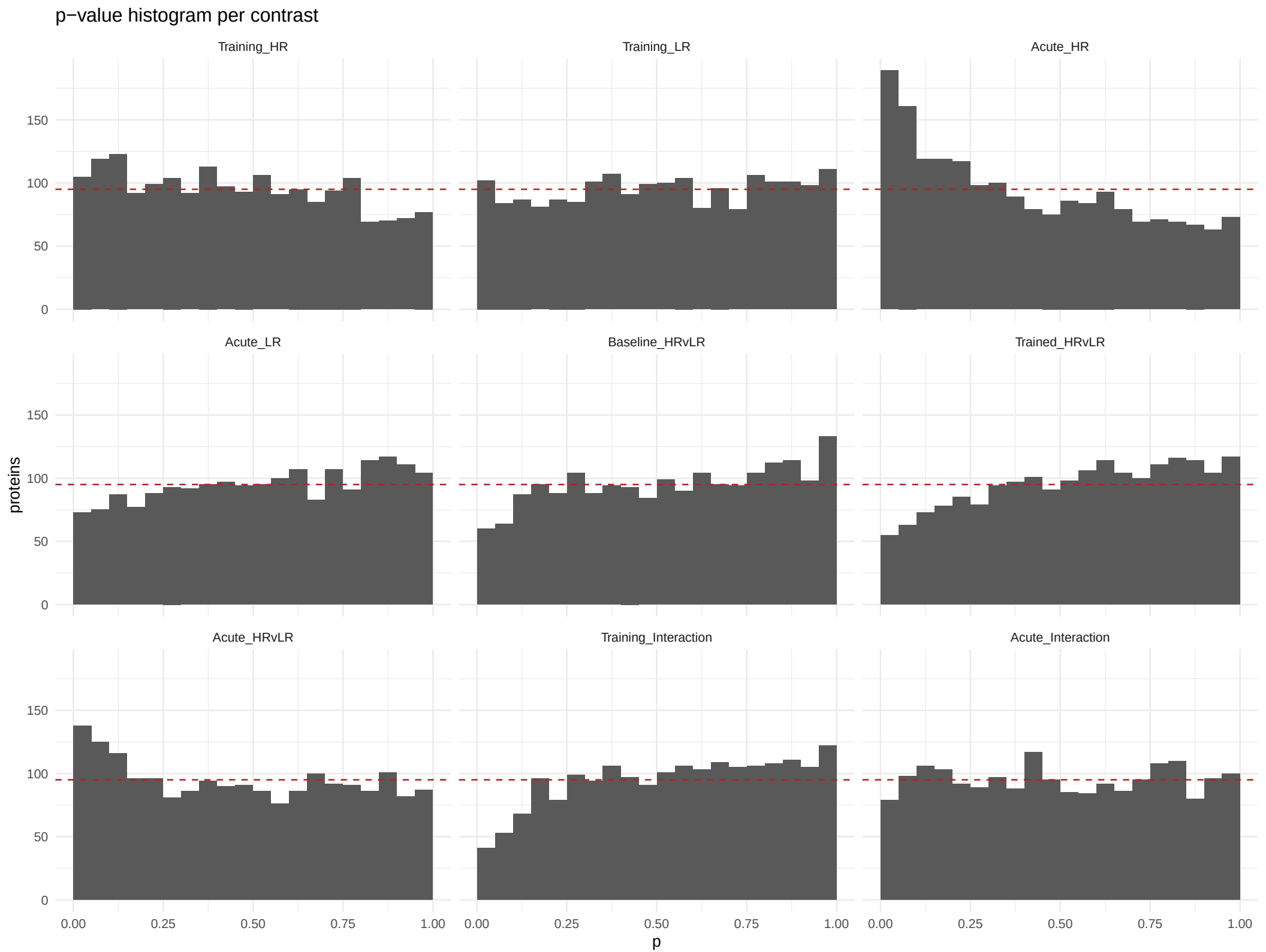


All 1,900 proteins by 45 biopsies, grouped by arm and time point; column labels are subjects. Fill: log2 abundance minus the protein's mean over biopsies. Colours stop at the 99th percentile of |value|, 1.9. Rows are ordered by average−linkage clustering (1 − Pearson r) of limpa's matrix, for display only. Grey: missing.

Contrast overview: proteins



All 1,900 proteins by the nine contrasts. Fill: log2 fold change from the limma fit; status: p and BH q within contrast. Colours stop at the 99th percentile of |value|, 0.91. Rows are ordered by average−linkage clustering (1 − Pearson r) of limpa's matrix, for display only. Grey: not tested. Table: c_data/protein_contrasts.xlsx, protein_by_contrast.

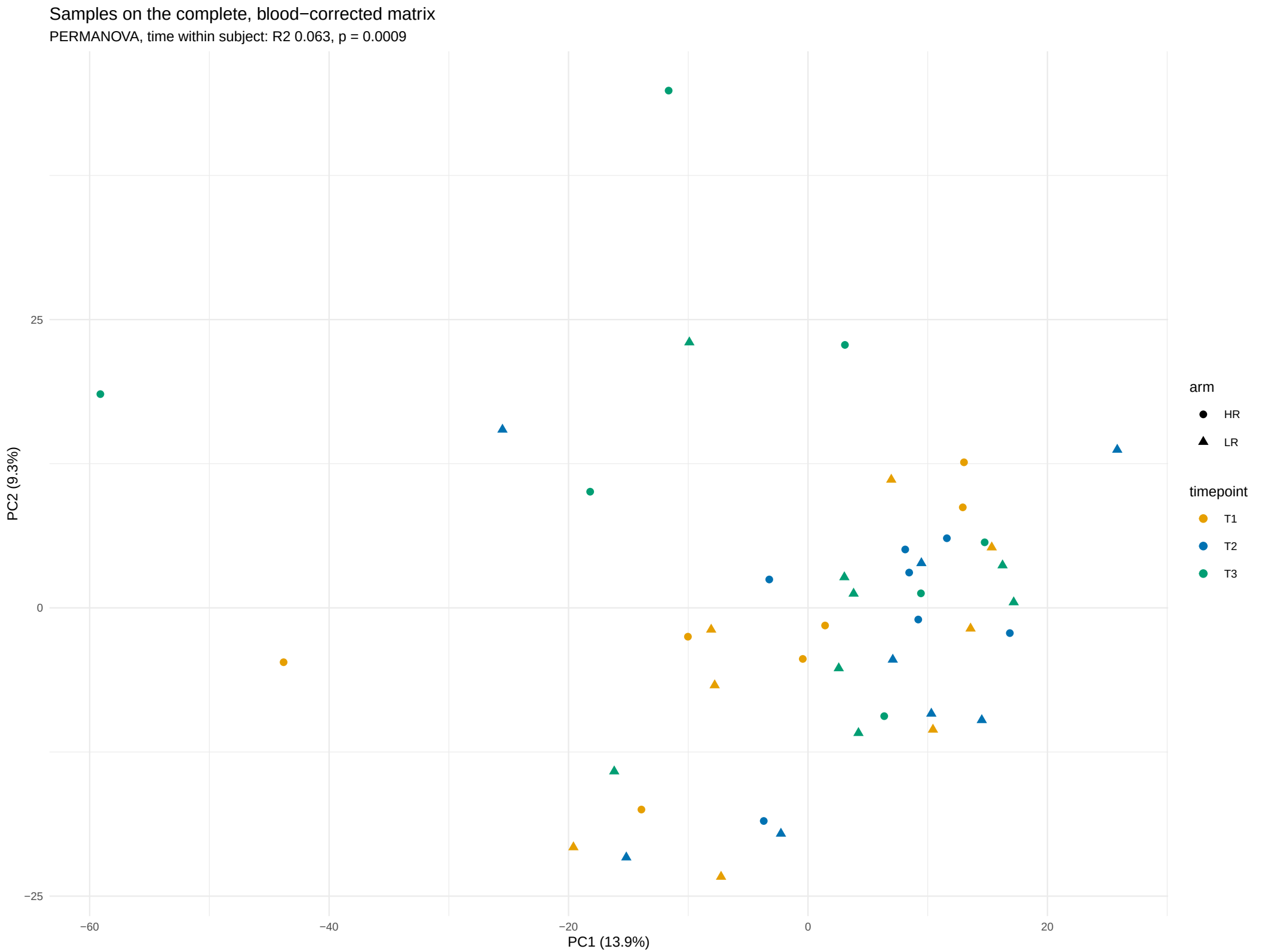

p−value histogram per contrast
Training_HR
Training_LR
Acute_HR
Acute_LR
Baseline_HRvLR
Trained_HRvLR
Acute_HRvLR
Training_Interaction
Acute_Interaction
proteins
p
0
50
100
150
0.00
0.25
0.50
0.75
1.00
Dashed line: a flat null (tested / 20). A spike at zero on a flat background is signal; a slope up toward 1 is a conservative test. Table: c_data/protein_contrasts.xlsx, protein_by_contrast.

Volcano per contrast

41 to 189 proteins at p < 0.05 per contrast; 0 at BH < 0.05 across all nine.



limma on limpa's quantification, subject blocked, blood as a covariate. Dashed line: p = 0.05. BH within contrast. Table: c_data/protein_contrasts.xlsx, protein_by_contrast.


Samples on the complete, blood−corrected matrix
PERMANOVA, time within subject: R2 0.063, p = 0.0009
PC2 (9.3%)
25
0
−25
−60
−40
−20
0
20
PC1 (13.9%)
arm
HR
LR
timepoint
T1
T2
T3
Proteins scaled to unit variance. Table: c_data/protein_contrasts.xlsx, pca, permanova.